CHAPTER 1

INTRODUCTION

1.1 Overview

Solar photovoltaic (PV) systems generate electrical energy by converting incident sunlight into electricity. The output of a PV panel depends on the available irradiance, operating temperature, electrical load and the angle at which light reaches its surface. A fixed panel is mechanically simple, but its orientation cannot follow the changing apparent position of the sun during the day. A dual-axis tracking mechanism attempts to adjust both horizontal and vertical orientation so that the panel can face the prevailing light direction.

An embedded solar tracker combines light sensors, a controller and electromechanical actuators to provide automatic adjustment. In the present project, an ESP32 DevKit V1 receives signals from four light-dependent resistors (LDRs) arranged in top, bottom, left and right directions. The controller determines the horizontal and vertical imbalance between the readings, applies filtering and deadband logic, and commands two MG996R servo motors. The servo-driven mount changes the orientation of the attached solar panel without requiring constant manual positioning.

The prototype also provides local monitoring through an INA219 current/power sensor, a DS18B20 digital temperature sensor and a 16 x 2 I2C LCD. An onboard Wi-Fi web server displays measurements in a browser and allows manual switching of a relay-connected DC load. The IP address is shown on the LCD to simplify browser access. The latest implemented firmware connects to a configured private Wi-Fi router; it does not create its own access point and does not require a separate web-dashboard login.

This project is presented as an engineering prototype for demonstrating automatic positioning, sensor acquisition, local user feedback and IoT-based control. Measurements recorded during testing also reveal a limitation: in strong outdoor ambient illumination, all four LDR ADC channels reached their maximum value of 4095, causing a programmed light-hold state. The report documents this observation openly and discusses required calibration rather than treating the outdoor tracking as fully validated.

1.2 PROBLEM STATEMENT

Conventional fixed photovoltaic modules cannot change their angle automatically as the sun moves across the sky. Manual repositioning is inconvenient, and a single-axis arrangement offers only one rotational degree of freedom. Low-cost tracking prototypes may also suffer from unstable movement when the light-sensor signals fluctuate, and they often lack live voltage, current or temperature monitoring. A further practical issue is that local demonstration systems can be inconvenient to operate when the user must connect a computer simply to view status or switch an auxiliary load.

The engineering problem addressed in this project is to construct a compact dual-axis PV-tracking prototype that autonomously responds to differences among four directional light sensors, limits unnecessary servo movements, presents useful measurements locally, and exposes a browser-based monitoring and manual control interface. The system must also recognize conditions in which the sensor readings cannot support a valid directional decision, such as complete ADC saturation.

1.3 OBJECTIVES

- To develop two-axis solar-panel orientation using four LDR sensors and an ESP32 controller.
- To implement sensor averaging, exponential filtering, proportional adjustment, hysteresis and direction confirmation to reduce servo hunting.
- To actuate horizontal and vertical movements using MG996R servo motors within configured mechanical limits.
- To measure PV-line voltage, current and power using the INA219 and panel/environment temperature using the DS18B20.
- To display live electrical values, sensor status, servo angles and the web-dashboard IP address on a 16 x 2 I2C LCD.
- To provide a local web dashboard with periodically refreshed measurements and manual relay ON/OFF control.
- To evaluate indoor light tracking and identify limitations under outdoor illumination using recorded serial-monitor results.

1.4 LITERATURE SURVEY

1.4.1 Two axes sun tracking system with PLC control [1]

Abdallah and Nijmeh developed an electromechanical two-axis tracking arrangement controlled using a programmable logic controller. Their study examined the collection of solar energy compared with a fixed reference surface, showing the value of tracking on both rotational axes. The present project adopts the same general aim of two-axis orientation but replaces the PLC architecture with a compact ESP32, analog LDR feedback and hobby servo actuators. Unlike an open-loop scheduled tracker, the current prototype uses light-sensor differences as the immediate orientation feedback.

1.4.2 A simple and low-cost active dual-axis solar tracker [2]

Hammoumi and co-authors described an active solar tracker using LDR sensors and two servomotors in a low-cost experimental arrangement. Their work is particularly relevant because it shows that directional irradiance sensing can be used for a mechanically simple two-axis prototype. The current design retains the inexpensive four-LDR/two-servo concept while adding signal filtering, saturation detection, electrical measurements, temperature sensing, an LCD and a local Wi-Fi dashboard. Performance improvements reported in that research are literature results and are not claimed for this particular prototype.

1.4.3 UV sensor-based dual-axis solar tracking [3]

Jamroen and colleagues investigated a tracking system using ultraviolet sensors and compared it with fixed and LDR-based configurations. Their study illustrates an alternative sensing strategy and highlights the dependence of tracking performance on sensor characteristics and environmental conditions. This consideration is directly relevant to the present prototype, since outdoor experiments showed that the installed LDR voltage dividers could become saturated. Alternative sensors, shading structures or redesigned dividers are potential future developments.

1.4.4 Current and temperature monitoring documentation [4], [5]

The Texas Instruments INA219 documentation describes a digital high-side voltage/current monitor that reports electrical quantities over I2C. The Analog Devices DS18B20 documentation describes a digital temperature sensor using the 1-Wire interface. These component references inform the electrical-monitoring subsystem: the INA219 is placed in series with the positive PV path, and the DS18B20 performs temperature conversion independently of the LDR tracking updates. Electrical measurements are interpreted according to the actual sensor placement.

1.5 EXISTING AND PROPOSED SYSTEM SUMMARY

A basic fixed photovoltaic arrangement is mechanically inexpensive but cannot adapt its orientation. A conventional light-seeking solar tracker adds feedback and actuation, but may offer no remote observability, no operator control of the output load and limited protection against noisy readings. Proposed or published tracking systems vary considerably in complexity, sensing technology, accuracy and energy requirements.

The implemented system integrates automatic positioning with a dedicated measurement and communications layer. LDR feedback affects only the servo targets, while INA219 readings, temperature sampling, LCD refresh and HTTP requests are handled as separate tasks. The resulting design is suitable for controlled prototype demonstrations and incremental development. It is not claimed to provide maximum-power-point tracking, battery-charge control, grid connection or unattended long-term outdoor operation.

1.6 HARDWARE AND SOFTWARE COMPONENTS USED

Hardware component	Quantity / role
ESP32 DevKit V1	1 - central controller with ADC, PWM and Wi-Fi
LDR sensors + 10 kOhm resistors	4 each - differential light sensing; divider values subject to outdoor calibration
MG996R metal gear servos	2 - horizontal and vertical actuation
DS18B20	1 - digital temperature measurement
INA219 module	1 - PV-line voltage/current/power monitoring
16 x 2 LCD with I2C backpack	1 - local information and IP display
Single-channel relay module	1 - browser-controlled DC-load switching
Solar panels	2 x nominal 6 V; series-connected for prototype

Mechanical mount / base	Dual-axis panel support and mounting
Power conversion and protection	Regulated supplies, compatible charge controller/BMS as applicable

Software: Arduino IDE with ESP32 board support; Arduino C/C++; ESP32Servo, Wire, LiquidCrystal_I2C, Adafruit_INA219, OneWire, DallasTemperature, WiFi and WebServer libraries; HTML, CSS and JavaScript embedded in program flash for the local browser dashboard.

1.7 ORGANIZATION OF REPORT

Chapter 2 describes system design and operation with a functional block diagram. Chapter 3 explains the principal hardware modules. Chapter 4 presents the physical interfaces and electrical-connection design. Chapter 5 introduces the software environment and communication libraries. Chapter 6 discusses the implementation logic, flowchart, scheduling and control algorithm. Chapter 7 records the observed tests and sensor values. Chapter 8 gives the applications, advantages and limitations. Chapter 9 concludes the report and describes future scope, followed by references.

1.8 SUMMARY

The project combines inexpensive directional light sensing, dual-axis movement, local monitoring and browser-based load switching. The following chapters explain the resulting embedded architecture and distinguish functions established during prototype testing from those requiring further outdoor calibration and quantitative validation.

CHAPTER 2

SYSTEM DESIGN AND OPERATION

The system is organized around an ESP32 microcontroller that reads multiple inputs and updates separate output subsystems. Four analog LDR voltages provide an estimate of directional illumination. A DS18B20 measures temperature, and an INA219 connected on the PV-line positive path measures electrical characteristics. The ESP32 produces the two PWM signals needed for the servo motors while also updating the LCD and handling local network requests.

2.1 BLOCK DIAGRAM

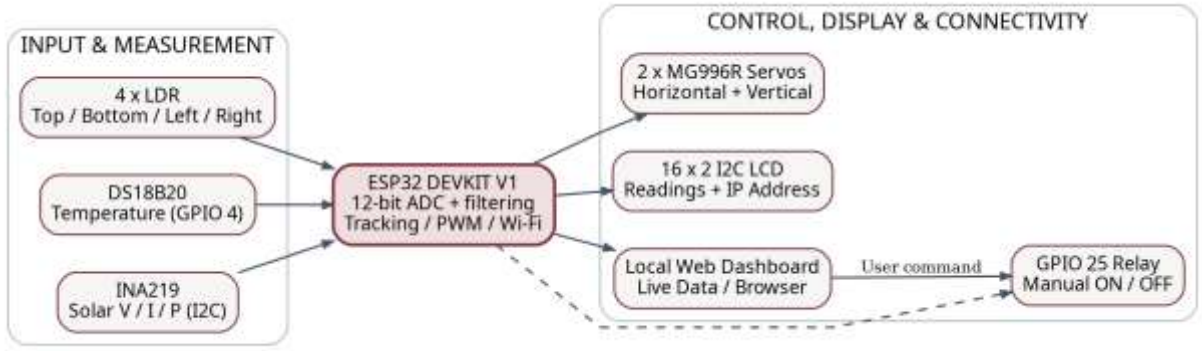


Figure 2.1 Functional block diagram of the ESP32 dual-axis solar tracking system

Figure 2.1 illustrates the separation of measurement, control and user interaction. The tracker can continue processing LDR values and moving the servos even while the router is unavailable. The browser dashboard accesses periodically refreshed data from the ESP32; manual browser commands update a dedicated relay output. The relay command does not change the LDR-based position-control algorithm.

2.2 WORKING MECHANISM

At startup, the ESP32 initializes the analog inputs, timer/PWM resources, DS18B20 conversion, I2C bus, LCD, INA219 and Wi-Fi stack. Each servo starts from a nominal 90-degree command within the mechanical range configured in the program. The main loop uses the millis() clock for independent periodic tasks. This is preferable to long blocking delays because electrical monitoring and web communication can take place while the tracker remains responsive.

For each directional measurement, 16 ADC samples are averaged and then combined with the previous value using an exponential smoothing filter. The control program computes a normalized horizontal difference from the left and right LDR values and a vertical difference from the top and bottom values. A deadband prevents adjustments when the light difference is small, and a larger threshold plus repeated confirmation is required to initiate or reverse movement. Commanded target angles are limited to the configured mechanical range.

The current implementation holds its position when all sensors are too dark or all four channels are at or above the saturation threshold. This behavior is intentional: when all readings are clipped at the ADC maximum, the calculated left/right and top/bottom differences contain no directional information. If sufficient valid light imbalance is detected, the commanded servo positions step toward newly calculated targets. Independent display tasks present measured voltage, current, power, temperature and servo status. The Wi-Fi dashboard provides a more detailed view, and a relay is switched manually through browser commands.

2.3 FUNCTIONAL DATA FLOW

Stage	Input	Processing / result
Sensing	4 x LDR analog values	16-sample mean and exponential smoothing
Decision	Left-right and top-bottom differences	Normalized error + thresholds + direction confirmation
Actuation	Target H/V angles	Two PWM servo commands, with angle limits
Electrical monitoring	INA219 I2C registers	Solar-line voltage, current and power
Environmental monitoring	DS18B20 1-Wire data	Temperature with non-blocking conversion
Local interface	Status variables	Two LCD pages and Wi-Fi IP overlay
Remote interface	Browser HTTP requests	JSON telemetry and GPIO25 relay ON/OFF

2.4 SYSTEM BOUNDARY AND CONTROL MODE

The tracker uses a local Wi-Fi router (station mode) rather than an ESP32-generated hotspot. The browser runs on a phone or computer connected to the same private network and accesses the IP address displayed on the LCD. No separate application or cloud service is required. In the latest firmware, the dashboard does not ask for a second username/password; consequently, local network access must be restricted by the Wi-Fi router. The relay is manually controlled, defaults to its commanded OFF state at boot and has no automatic battery-cutoff function.

The INA219 position is important for data interpretation. As installed on the positive line between the series solar panels and the solar charger input, it describes that PV-side path. It does not directly report the battery state-of-charge or battery terminal voltage. Voltage and current values should not be labeled battery measurements unless the wiring and measurement placement have been changed and checked.

CHAPTER 3
HARDWARE DESIGN AND DESCRIPTION

This chapter describes the hardware that forms the proposed embedded prototype. Selection criteria include cost, GPIO compatibility, current requirements, communication interface and mechanical suitability. The design intentionally uses separate power domains for logic electronics and higher-current servo actuation.

3.1 ESP32 DEVKIT V1

The ESP32 is a 32-bit microcontroller family integrating processing, GPIO resources, timers and 2.4 GHz Wi-Fi. The DevKit board provides accessible headers and USB programming. Four ADC1-capable inputs, GPIO34, GPIO35, GPIO32 and GPIO33, are assigned to the four LDR voltage dividers. The current firmware configures 12-bit analog readings, giving nominal digital values from 0 through 4095. This pin selection avoids relying on ADC2 for LDR acquisition while Wi-Fi is running [6].

The ESP32 also generates the servo-control pulse signals at GPIO13 and GPIO14, communicates on the I2C bus using GPIO21 (SDA) and GPIO22 (SCL), reads the DS18B20 at GPIO4 and drives the relay module at GPIO25. Firmware tasks update sensors, actuators and the web server independently within the main loop. A regulated board supply and common control reference are required for stable operation.

3.2 LIGHT DEPENDENT RESISTORS (LDRs)

An LDR is a resistive light sensor whose resistance varies with incident illumination. For each sensor, the design places the LDR on the 3.3 V side of a divider and a fixed nominal 10 kOhm resistor on the ground side. The voltage at their junction is measured by an ESP32 ADC pin. With this orientation, increased light generally decreases LDR resistance and increases the ADC voltage.

The four devices are mounted as TOP, BOTTOM, LEFT and RIGHT to detect directional light imbalance. When the sunlight is strongly directional, a small cross-shaped shading barrier between the LDRs can increase differences among channels. Under bright open-sky conditions, however, low LDR resistances may push all four divider outputs toward the ADC ceiling. In the observed outdoor test, this happened even in shade, motivating a future resistor-divider and optical-shading revision.

3.3 MG996R SERVO MOTORS

Two MG996R metal-gear hobby servos rotate the panel assembly on orthogonal axes. The horizontal servo changes azimuth-like orientation and the vertical servo changes elevation-like orientation. The code drives both channels at 50 Hz, with calibrated pulse limits set to 1000-2000 microseconds in the current project. Servo commands are restricted to the configured 30-180 degree software range.

Exact usable angles depend on the mechanical mount and must be verified to prevent collision or overload.

MG996R servos can draw substantially more current during startup, rapid movement or stall than the ESP32 supply can deliver. Therefore, their motor power must come from an independent regulated 5-6 V supply with suitable current capacity. A common low-voltage ground reference is required for the control signals. Connecting the servo power wires directly to the solar panel or a nominal 12 V battery would be unsafe.

3.4 DS18B20 DIGITAL TEMPERATURE SENSOR

The DS18B20 measures temperature digitally via the 1-Wire protocol. The sensor data line connects to ESP32 GPIO4 with an appropriate 4.7 kOhm pull-up to the 3.3 V logic rail unless an equivalent pull-up is present on the sensor module. The firmware operates the device at 12-bit resolution, for which the specified conversion time is up to 750 ms [5].

Instead of waiting for a conversion inside a blocking delay, the code requests a conversion, continues ordinary tracking tasks and later retrieves the completed value. Measured temperature is displayed on the LCD and browser dashboard and included in the serial diagnostic output. The data indicates the temperature at the physical sensor location; it should not automatically be interpreted as cell-junction temperature inside the PV panel.

3.5 INA219 VOLTAGE, CURRENT AND POWER MONITOR

The INA219 is a digital shunt and bus-voltage monitor with an I2C-compatible interface. A common breakout module is used at address 0x40. It provides shunt voltage, bus voltage, current and calculated power. The source-side voltage in the code is obtained by adding the shunt voltage drop to the measured load-side bus voltage. This means that module wiring directly determines what is being measured [4].

For the implemented arrangement, VIN+ is connected toward the combined positive of the series solar panels and VIN- continues toward the charge controller PV positive input. The sensor logic VCC operates from a suitable supply and shares I2C SDA/SCL with the LCD. The device has a 26 V common-mode bus range, but the allowable current of a particular breakout depends on its shunt resistor and PCB construction. The installed system must be checked against those ratings before operation.

3.6 16 x 2 I2C LCD DISPLAY

The LCD provides 16 characters on each of two lines. Its I2C backpack reduces the number of required ESP32 pins and is discovered at commonly used addresses 0x27 or 0x3F. In normal operation, the firmware alternates between an electrical/temperature page and a tracking/status page. It updates only changed characters to minimize bus traffic and visible flicker.

While Wi-Fi is connecting, the LCD displays a connection message. Once the router connection is ready, the assigned IP address remains visible until the user opens the dashboard. Afterwards, the normal measurement pages resume and the IP reminder returns periodically. Some 5 V LCD backpacks use I2C pull-ups to 5 V; the ESP32 requires compatible 3.3 V signal levels or an appropriate bidirectional level shifter.

3.7 SINGLE-CHANNEL RELAY MODULE

A one-channel relay module allows the controller to switch a small external DC load, such as a compatible LED lamp or fan. The control input is attached to GPIO25. The most recent program configures the relay as active-HIGH, where a HIGH output requests the ON state and LOW requests OFF. The actual module input polarity and contact action must be verified before the final load is attached. The command state is reflected in dashboard telemetry.

Use the relay COM and NO terminals to arrange for the load to be disconnected when the contacts are released. The relay supply must match the module rating, commonly 5 V, and a 3.3 V GPIO must be sufficient to operate the input interface. The relay does not act as a battery management system; suitable over-current, over-discharge and charging protections remain external responsibilities.

3.8 SOLAR PANELS, CHARGE CONTROLLER AND MECHANICAL ASSEMBLY

The prototype uses two nominal 6 V solar panels connected in series to form the available source for the system. A measured combined value of about 13.5 V was previously observed. The correct series connection joins the positive lead of one panel to the negative lead of the other, leaving two free conductors as the combined positive and negative terminals. Whether such an arrangement can charge a particular nominal 12 V battery depends on the loaded panel voltage, charge-controller type and battery chemistry.

The solar-panel mount must provide adequate stiffness, balanced weight distribution and sufficient clearance for both rotation axes. A suitable solar charge controller and battery/BMS are required if battery charging is included. The exact controller terminal layout and battery type were not supplied for this report, so the power-path diagram in Chapter 4 is illustrative rather than a claim of verified charger wiring.

3.9 HARDWARE SPECIFICATION SUMMARY

Subsystem	Implemented interface / configuration
LDR array	Four 10 kOhm dividers into GPIO34, 35, 32, 33
Servo motor H / V	GPIO13 / GPIO14; 50 Hz; configured 30-180 degrees
Temperature	DS18B20 on GPIO4; 12-bit non-blocking conversion
Electrical	INA219 on I2C address 0x40
LCD	16 x 2 display at 0x27 or 0x3F
Wireless	ESP32 2.4 GHz station mode, local router
Web server	Local HTTP port 80, browser telemetry/control

Relay	GPIO25; active-HIGH in latest source
Battery cutoff	Not implemented in current firmware

CHAPTER 4

HARDWARE IMPLEMENTATION

Hardware implementation brings together the sensing, control, actuation and monitoring modules using the assigned GPIOs and external power paths. Separate attention is required for the LDR analog voltages, I2C signal-level compatibility, high-current servo supply and the solar/battery power circuit. The following diagrams are functional wiring illustrations generated from the current firmware and available project description; they are not photographs of the final assembled hardware.

4.1 CIRCUIT CONNECTION DIAGRAM

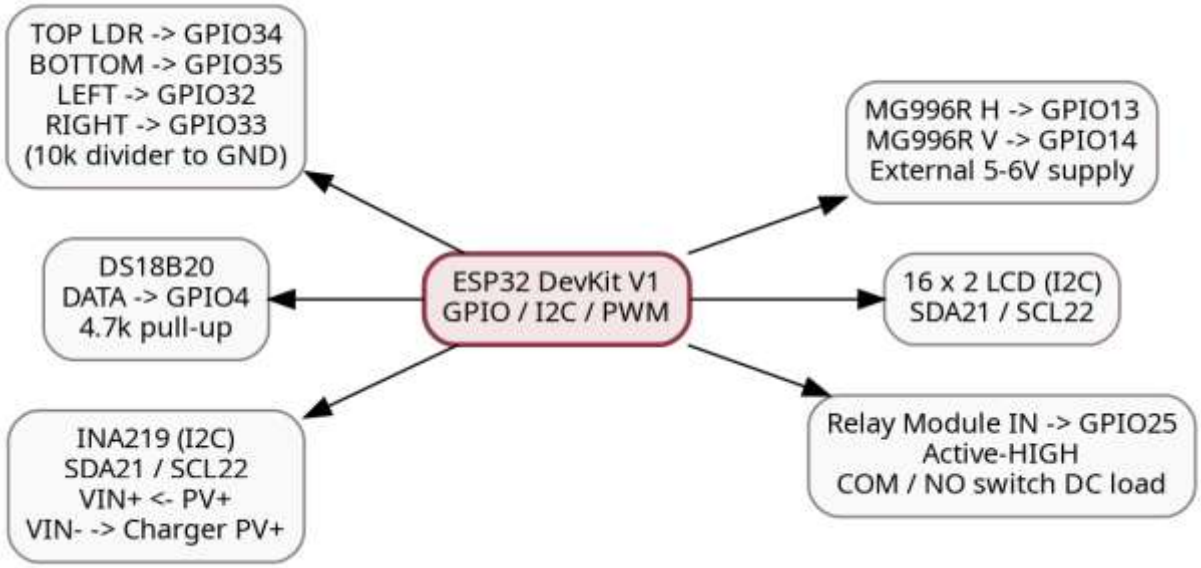


Figure 4.1 Signal connection diagram for the main ESP32 hardware

4.2 PIN CONNECTIONS AND INTERFACE EXPLANATION

ESP32 pin	Connected unit	Signal / purpose
GPIO34	TOP LDR	Analog voltage divider output
GPIO35	BOTTOM LDR	Analog voltage divider output
GPIO32	LEFT LDR	Analog voltage divider output
GPIO33	RIGHT LDR	Analog voltage divider output
GPIO13	Horizontal MG996R	Servo control PWM signal
GPIO14	Vertical MG996R	Servo control PWM signal
GPIO4	DS18B20 DQ	Digital 1-Wire data; pull-up needed
GPIO21	LCD + INA219	I2C SDA
GPIO22	LCD + INA219	I2C SCL
GPIO25	Relay IN	Active-HIGH manual switching

All four LDR dividers use 3.3 V and GND as reference rails. The present circuit assumes an LDR tied to the 3.3 V rail and a fixed 10 kOhm resistor from the analog input to ground. The two servos share a common reference ground with the ESP32 but receive their operating power from an external regulated high-current supply. For the LCD and INA219, both devices use GPIO21 and GPIO22 because I2C is a shared addressed bus.

INA219 VIN+ and VIN- are series measurement terminals, not substitute power inputs. They should carry the positive PV path whose current is being measured. The solar-negative/charge-controller grounding arrangement must follow the manufacturer guidance, especially if the power converter has isolated or switched negative terminals.

4.3 CONCEPTUAL POWER SUPPLY ARRANGEMENT

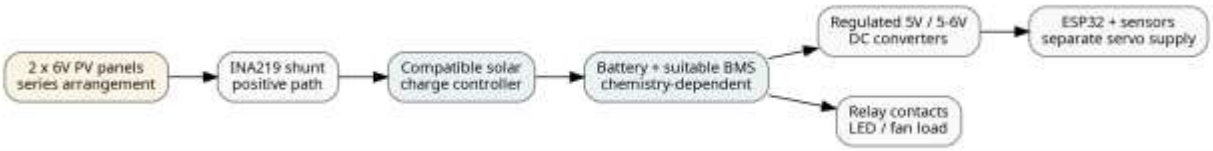


Figure 4.2 Conceptual panel, charger, battery, regulator and load power path (verification required)

Figure 4.2 identifies the intended functional order of energy conversion. It does not assert that a particular battery, BMS or charger has been tested or approved. A regulator supplies the ESP32 with an appropriate stable voltage; an independent high-current 5-6 V regulator supplies the MG996R servos. The relay contacts switch a separately protected load circuit. Before applying power, all actual terminals and polarity must be checked with the specific charge-controller and battery manuals.

4.4 ASSEMBLY PROCEDURE

1. Fix the horizontal and vertical MG996R servos to the mounting structure and verify motion clearance by hand with power removed.
2. Attach the panel and mount the four LDRs in top, bottom, left and right positions with comparable exposure to light.
3. Construct the four LDR divider networks and connect their midpoints to GPIO34, GPIO35, GPIO32 and GPIO33.
4. Wire both servo signal inputs, the DS18B20 data/pull-up and the separate servo supply with common low-voltage ground.
5. Connect the I2C LCD and INA219 to GPIO21 and GPIO22, checking their module supply and I2C pull-up levels.
6. Connect the relay-control input to GPIO25 and verify module polarity using a low-power test before connecting the actual load.
7. Connect solar panels and the charge-controller wiring according to the equipment labels; place INA219 VIN+/VIN- in the intended positive PV measurement path.

8. Upload firmware, confirm the I2C scan, monitor raw LDR values, verify mechanical directions and finally test the private-network dashboard.

4.5 HARDWARE SAFETY CONSIDERATIONS

The ESP32, servos and LCD logic are low-voltage electronic devices; the PV source and battery energy path must not be connected directly to logic pins. Servo power demands can produce substantial voltage drop, resets or overheating if undersized wires or regulators are used. Battery-powered loads require a suitable BMS when applicable and a fuse matched to expected current and wiring. The optional relay is not an automatic over-discharge device in the current configuration.

When operating outdoors, avoid water ingress and direct solar heating of unprotected electronics. A full sunlight commissioning procedure should also test optical sensor saturation and panel movement under mechanical loading. No formal environmental or electrical certification was carried out in the data available for this report.

CHAPTER 5

SOFTWARE DESCRIPTION

The firmware is written in Arduino-compatible C++ and targets the ESP32 DevKit. The architecture combines short periodic sensor/control tasks with display and network handling. It is not an operating-system-based implementation and does not depend on a remote server or cloud subscription.

5.1 ARDUINO IDE

Arduino IDE provides a source editor, board/package manager, compiler, upload tools and Serial Monitor. After installing the supported ESP32 board platform, the project sketch is compiled for the ESP32 Dev Module or the exact matching target board. Serial Monitor is configured to 115200 baud to view raw LDR readings, directional error, servo positions, DS18B20 temperature, INA219 values and light tracking status.

The firmware begins by initializing input modes, PWM outputs, I2C devices and the Wi-Fi station. The `setup()` function performs one-time initialization while `loop()` schedules periodic functions. The code maintains a non-blocking DS18B20 conversion and does not wait indefinitely for Wi-Fi before starting tracking.

5.2 REQUIRED LIBRARIES

Library	Purpose in project
Arduino.h	Core GPIO, ADC, time and serial functions
ESP32Servo.h	PWM servo attach and position commands
Wire.h	I2C master bus and address scan
Adafruit_INA219.h	INA219 electrical registers and calibrated current/power
LiquidCrystal_I2C.h	16 x 2 I2C LCD control
OneWire.h	Physical 1-Wire bus
DallasTemperature.h	DS18B20 temperature conversion and validation
WiFi.h	ESP32 connection to private 2.4 GHz Wi-Fi router
WebServer.h	Local HTTP routes for page, JSON and relay commands

5.3 PROGRAMMING AND COMMUNICATION INTERFACES

The LDR readings use analog-to-digital conversion. The calculated control outputs are transmitted to the servos as pulse-width commands. The DS18B20 uses a single-wire digital bus, while the INA219 and LCD share I2C. Wi-Fi and HTTP provide connectivity to the browser. The current web implementation serves a local HTML/CSS/JavaScript page and provides JSON telemetry through the `/api/data` endpoint. A POST request to `/api/load` with `mode=on` or `mode=off` changes the relay command.

The telemetry contains solar voltage, solar current, solar power, temperature, tracking status, servo angles, four LDR values, normalized errors and the relay command state. The HTML page refreshes this information periodically. Measurements from missing or invalid sensors are presented as unavailable rather than implying a valid zero-value measurement.

5.4 WI-FI AND LCD IP DISPLAY

The current firmware operates in station-only mode. It connects to one configured router and does not start the earlier prototype access point named SolarTracker-Setup. If the connection is unsuccessful, the LCD displays a Wi-Fi-not-ready message. Once connected, it displays the address assigned by the router until the dashboard is opened. After opening, the LCD returns to normal information pages and periodically repeats the IP address as a reminder.

The dashboard currently has no additional HTTP authentication. This was chosen for local demonstration convenience but means that a user on the same private network can open the page and send relay control commands. For deployment, the interface should include authentication or another appropriate access-control mechanism, and it should never be forwarded directly to the public internet.

5.5 DATA VALIDATION AND DIAGNOSTICS

The firmware checks whether I2C devices respond during initialization and validates DS18B20 temperature readings before presenting them. Analog LDR values are smoothed to limit noise-induced motion. On the control side, low average light or simultaneous saturation of all four sensors disables directional movement and retains the present servo position. The tracking status reported to the user is therefore one of the operating indications STABLE, TRACKING or LIGHT HOLD.

The availability of Serial Monitor diagnostics was important in identifying the actual outdoor issue. Rather than assuming an actuator failure, the displayed raw LDR data showed that every channel was fixed at its full-scale ADC value, explaining why the normalized error became zero and why the code entered the hold condition.

CHAPTER 6

SOFTWARE IMPLEMENTATION

The software continuously repeats a set of short tasks, each controlled by a scheduled interval. The design separates sampling and filtering from the desired-angle computation and separates desired-angle computation from the incremental servo movement. This structure makes the motion response less sensitive to brief sensor disturbances and leaves time for temperature, energy monitoring and network communication.

6.1 FLOWCHART

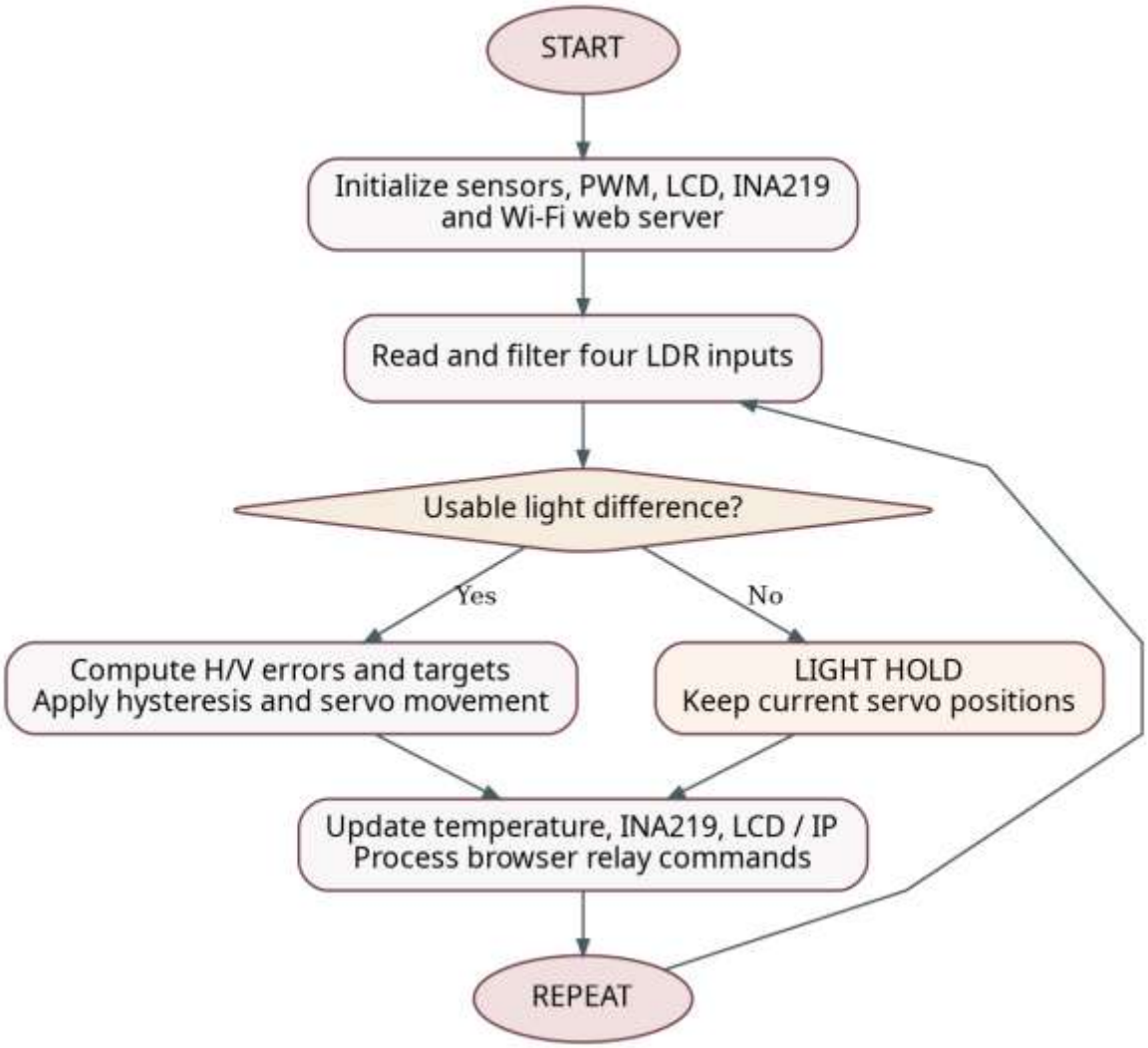


Figure 6.1 Control and monitoring flowchart of the implemented ESP32 system

6.2 TRACKING ALGORITHM

1. Initialize the four analog channels and servo pulse outputs; center the two servo commands at 90 degrees.

2. Read sixteen ADC samples from each LDR and calculate the average for each channel.
3. Apply exponential smoothing to each channel using the configured filter coefficient.
4. Calculate mean light level and test the low-light and four-channel saturation conditions.
5. If illumination is invalid, reset movement-direction states and retain the present servo commands.
6. Otherwise, calculate horizontal and vertical normalized percentage errors.
7. Apply stop threshold, start threshold and direction-confirmation logic to both axes.
8. If motion is authorized, adjust the desired target proportional to error, within the target-step and mechanical limits.
9. At the servo task interval, increment the commanded servo positions toward their desired targets.
10. Independently update temperature, INA219 readings, LCD pages, local IP overlay, serial diagnostics and the HTTP server.
11. Accept browser relay ON/OFF messages and update the configured GPIO25 output.
12. Repeat until the hardware is powered off.

6.3 MATHEMATICAL MODEL

Let L, R, T and B denote the smoothed ADC measurements from the left, right, top and bottom LDRs. The normalized axis errors used by the firmware are:

```
E_H = 100 x (L - R) / (L + R)
E_V = 100 x (T - B) / (T + B)
```

If the denominator for a pair is too small, the code returns zero for that axis. The smoothed measurement is computed as:

```
Filtered(n) = Filtered(n-1) + alpha x [Raw(n) - Filtered(n-1)]
alpha = 0.18
```

Two thresholds control the response. An existing movement is stopped when the absolute normalized error is at or below 3%, while a new movement must satisfy an error of at least 7% and direction confirmation across three assessments. The proportional target adjustment uses a gain of 0.18 and a maximum target-angle change of 7 degrees per update. Actual servo commands move toward the target by at most 3 degrees at each servo task. Direction signs are configured to suit the mechanical mounting.

6.4 CONTROL AND TIMING PARAMETERS

Parameter	Value	Function
ADC resolution	12 bit (0-4095)	Sensor digitization
ADC samples	16 per sensor	Short-term averaging
Filter alpha	0.18	Exponential smoothing
Start / stop error	7% / 3%	Hysteresis
Direction confirmation	3 cycles	Avoid frequent reversals
Proportional gain KP	0.18	Target movement response

Max target change	7 degrees	Limits target jump
Servo step	3 degrees	Limits each command increment
Servo bounds	30-180 degrees	Software mechanical envelope
Sensor task	60 ms	LDR acquisition
Tracking task	100 ms	Desired-angle update
Servo task	15 ms	Incremental servo command
INA219 task	1000 ms	Electrical refresh
LCD task	1000 ms	Display refresh
Serial task	500 ms	Diagnostic output
DS18B20 conversion	750 ms	Non-blocking 12-bit conversion
Light saturation cutoff	4050 (all four)	LIGHT HOLD trigger

6.5 LIGHT HOLD AND SATURATION PROTECTION

The light-valid condition is false if the mean sensor signal is below 120 or if all four channels are at or above 4050. In either case, the tracker holds its current position; the code does not move the panel based on unavailable or contradictory directional information. This condition protects against meaningless movement decisions, but it also exposes a practical shortcoming of the original LDR divider design under strong ambient daylight.

The observed log `TOP:4095 BOTTOM:4095 LEFT:4095 RIGHT:4095` produces `ERR_H:0.0 ERR_V:0.0`. Because the values have reached the ADC maximum, it is impossible for the algorithm to determine which side is brighter. Lowering or removing the software saturation threshold alone would not recover the missing sensor information. The hardware voltage-divider and directional shading arrangement must be tuned and retested outdoors.

6.6 MANUAL RELAY CONTROL

The dashboard sends a POST request specifying the desired state. The handler recognizes only ON and OFF, calls the relay-control function and returns a text acknowledgement. In the current code, `RELAY_ACTIVE_LOW = false`, so the selected relay module is treated as active-HIGH. The actual LED/fan load must also use the correct normally-open contact wiring to ensure that the physical load state agrees with the browser command.

Automatic battery-voltage monitoring, auto load disconnect and autonomous recharge detection were removed from this firmware after the user decided to retain browser-only control. Therefore, the relay must be described as a manual load switch, not a battery-protection controller.

CHAPTER 7
EXPERIMENTAL RESULTS

The solar tracking prototype was tested during development using indoor artificial illumination and subsequently outdoors. The indoor demonstrations established observable movement response under directional light and supported integration of the LCD, temperature sensor, electrical monitor and Wi-Fi interface. Numerical serial logs supplied for outdoor tests allow a more specific assessment of sensor behavior. This section separates these observed facts from unmeasured performance claims.

7.1 INDOOR LIGHT TRACKING TEST

When the project was exposed to a room bulb or directional light entering the room, the user observed horizontal and vertical movement of the solar panel as the direction of the light changed. The two servos were able to respond to directional sensor differences, and the system could settle into its stable state when the imbalance was reduced. This is a qualitative functional observation. No calibrated angular tracking error, settling-time series or repeated-cycle statistics were supplied, so the report does not assign numerical accuracy or repeatability to indoor operation.

7.2 OUTDOOR SHADE TEST

On moving the system outdoors into an area that was not in direct sunlight, the serial monitor showed all four LDR channels at the maximum 12-bit reading. The tracker remained at horizontal command H = 80 degrees and vertical command V = 122 degrees with no new target movement. The temperature sensor reported values around 23.75-23.87 degrees Celsius during the shown series; INA219 voltage was approximately 11.66-11.67 V, while indicated current ranged from 0.3 to 0.9 mA in those excerpts.

```
TOP:4095 BOTTOM:4095 LEFT:4095 RIGHT:4095
ERR_H:0.0 ERR_V:0.0 | H:80 V:122 | LIGHT HOLD
```

Recorded serial-monitor excerpt, outdoor shade test (shortened only for legibility)

7.3 DIRECT SUNLIGHT TEST

The separate direct-sunlight log again showed TOP, BOTTOM, LEFT and RIGHT equal to 4095. The normalized errors remained zero, the servo commands remained H = 80 degrees and V = 122 degrees, and the status remained LIGHT HOLD. The displayed DS18B20 temperature varied approximately from 26.06 to 26.56 degrees Celsius in this sequence; INA219 voltage was 11.72 V and current values were approximately 0.9-1.2 mA.

```
TOP:4095 BOTTOM:4095 LEFT:4095 RIGHT:4095
ERR_H:0.0 ERR_V:0.0 | H:80 V:122 | TEMP:26.56 C
INA219: V:11.72V I:1.0mA P:0.01W | LIGHT HOLD
```

Recorded serial-monitor excerpt, direct sunlight test (reformatted to two lines)

7.4 EXPERIMENTAL OBSERVATION TABLE

Test condition	Observed result	Interpretation
Indoor bulb / directional light	Two-axis movement was observed	Functional light-seeking under indoor conditions
Outdoor in shade	All four LDRs = 4095; LIGHT HOLD	ADC saturation / no directional contrast
Outdoor direct sun	All four LDRs = 4095; LIGHT HOLD	ADC saturation persists in brighter condition
Servo position in outdoor logs	H=80 degrees; V=122 degrees	Position retained by hold logic
Temperature in shown tests	About 23.75-26.56 C	Digital temperature measurement present
INA219 bus/source voltage	About 11.66-11.72 V	PV-side monitored voltage in shown sequence
INA219 current	About 0.3-1.2 mA	Very low measured current; power approx. 0.00-0.01 W
Web dashboard / relay	Reported operational during project development	Manual LAN control; no formal response-time study

7.5 ANALYSIS OF LDR SENSOR SATURATION

The full-scale 4095 reading on every LDR channel is consistent with ADC saturation. In the divider orientation used by the prototype, intense incident light can reduce each LDR resistance until the ADC input approaches the logic rail. Once the ADC output is clipped, the recorded value cannot indicate further differences in incident brightness. Outdoor shade still contains significant diffuse sky illumination and therefore does not necessarily prevent clipping.

A controlled verification procedure is to fully cover one LDR with an opaque cap and observe whether its ADC value falls sharply. If the channel stays at 4095 when fully covered, wiring, GPIO setup or a sensor fault should also be checked. If the channel recovers correctly under complete darkness, the likely next steps are to test lower ground-side divider resistances (such as 2.2 kOhm followed by 1 kOhm as appropriate), add a shading cross or optical attenuator and repeat outdoor measurements. These are proposed corrective experiments; they were not reported as successfully completed.

7.6 ELECTRICAL MONITORING INTERPRETATION

The INA219 reports voltage, current and power at the point where it is installed. For the provided outdoor log, multiplying 11.72 V by 1.0 mA yields approximately 0.01172 W, corresponding to the printed 0.01 W. This confirms numerical consistency of the displayed units but does not establish the PV module's rated capability. Such low currents may result when the charger or load is drawing very little current or when the sensor is not installed on the intended loaded path.

Because the provided measurements do not contain an independent calibrated instrument comparison or known resistive-load test, the report does not conclude that the panel successfully charged a

battery, produced a specific daily energy yield or improved efficiency compared with a stationary panel.

7.7 RESULTS DISCUSSION AND LIMITATIONS

The combined results show that the system has useful prototype functionality under indoor lighting and that its control firmware correctly avoids action when sensor information becomes unusable. However, dependable outdoor tracking is a remaining development objective. The source code and user demonstration history support the existence of live measurement and browser relay control, but no laboratory report of browser latency, sustained communication uptime, relay-contact endurance or battery discharge tests is available.

The measured values presented here are actual examples from the user-provided serial logs, not averages over a statistically designed experiment. The strongest justified conclusion is that the system architecture works as an integrated prototype, with outdoor LDR saturation identified and requiring remediation before claims of continuous sunlight tracking or energy gain can be made.

CHAPTER 8

APPLICATIONS, ADVANTAGES AND DISADVANTAGES

8.1 APPLICATIONS

- Academic demonstration of analog sensor interfacing, servo-control algorithms and low-cost photovoltaic orientation.
- Embedded systems laboratories illustrating periodic scheduling, filtered measurements, I2C and 1-Wire integration.
- IoT teaching projects showing local HTTP dashboards and browser-operated DC relay switching.
- Small research prototypes for comparing optical tracker geometries, shading barriers and sensor voltage-divider designs.
- Expandable PV-monitoring platforms for controlled demonstrations with appropriately rated modules and protective power equipment.

8.2 ADVANTAGES

- Achieves two-axis controlled movement with four inexpensive photoresistors and two readily available servo motors.
- Uses proportional motion, hysteresis, filtering and direction confirmation to reduce needless oscillation during valid sensing.
- Integrates local LCD feedback with a browser dashboard using the ESP32 Wi-Fi capability.
- Monitors PV-side voltage, current, calculated power and digital temperature in a single controller.
- Provides manual ON/OFF control of a DC load without requiring a custom mobile application.
- Preserves light-tracking tasks independently of Wi-Fi connection establishment.
- Includes a deliberate LIGHT HOLD state that makes sensor saturation visible rather than silently commanding arbitrary movement.

8.3 DISADVANTAGES AND LIMITATIONS

- Outdoor light saturation: the installed four LDR channels reached 4095 in the supplied shade and direct-sun test cases.
- Tracking accuracy and energy gain have not been quantified against a calibrated reference or a fixed PV panel.
- The ESP32 GPIO supply is insufficient to power two MG996R servos, so additional regulated power hardware is necessary.
- The current dashboard requires access to the configured Wi-Fi network and does not operate as a standalone public hotspot.

- The simplified local dashboard has no separate login, so network isolation is important for relay-control security.
- The relay is manual only: no automatic battery protection or state-of-charge sensing is implemented.
- PV-line current readings of around 0.3-1.2 mA do not validate meaningful charger or load power delivery.
- The prototype has not been validated for long-term outdoor exposure, waterproofing, mechanical wind loading or industrial safety.

8.4 ENGINEERING TRADE-OFFS

The project prioritizes accessibility, modular construction and ease of demonstration over industrial accuracy or autonomous operation. Optical LDR tracking is low-cost but requires careful exposure and resistor calibration across changing illumination. Servo positioning is straightforward but may be unsuitable for large panel loads or harsh outdoor conditions. Local Wi-Fi provides a simple, fast user interface but is not a substitute for secured remote connectivity. These trade-offs should guide future modifications.

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 CONCLUSION

An ESP32-based dual-axis solar-tracking prototype was implemented using four LDR sensors, two MG996R servo motors, an INA219 electrical monitor, a DS18B20 temperature sensor, a 16 x 2 I2C LCD, local Wi-Fi dashboard and manually switched relay output. The controller processes normalized light imbalances using filtered sensor data and motion hysteresis, while independent monitoring tasks update the display and serve browser requests.

Indoor testing and project demonstrations established observable servo movement under directional light, working temperature/electrical monitoring and convenient local web-based control. The LCD-based IP display allows a phone on the same Wi-Fi network to open the dashboard without the earlier separate ESP32 hotspot or web-login step. The relay remains manually controlled; battery-protection automation is intentionally outside the current firmware.

The outdoor experiment identified a significant and well-defined limitation. All four LDR ADC inputs reached 4095 both in outdoor shade and direct sunlight, causing normalized directional errors to remain zero and activating LIGHT HOLD. The result indicates a need for improved LDR divider calibration and optical direction sensing before outdoor solar tracking can be considered validated. No fixed-versus-tracking energy-gain result can responsibly be claimed from the data presented.

Overall, the project demonstrates an effective educational integration of light sensing, embedded actuation, electrical measurement and IoT user interfaces, while also showing the value of real test data in diagnosing system limitations.

9.2 FUTURE SCOPE

1. Optimize each LDR divider for bright outdoor light using controlled resistance sweeps and calibrated irradiance test conditions.
2. Add an opaque cross-shaped shade or optical attenuating tube around the sensor array to improve directional contrast.
3. Validate full sunlight operation across morning, midday and late afternoon with logged raw ADC readings and actual servo positions.
4. Use sensor fusion or a solar-position model when optical readings are invalid or saturated.
5. Characterize tracking accuracy, settling time, servo power consumption and mechanical backlash under realistic mounting loads.
6. Add calibrated PV energy logging and compare measured Wh production against a fixed-reference panel over equivalent conditions.
7. Improve weatherproofing, structural mounting, cable routing and safe operation under wind and rain.

8. Add network authentication and TLS through an appropriate gateway if access extends beyond a trusted private LAN.
9. Integrate a suitable battery monitor and hardware BMS/charge controller before considering automatic load cutoff.
10. Consider dedicated low-power actuators and position feedback for a robust, larger-scale solar-tracking system.

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